

Vehicle-to-Grid (V2G)

Understanding the problem space: technology, markets, regulation, and data

The Opportunity: Why the Timing Matters

V2G is not a solved market. It is a market in formation. The regulation is converging, the hardware is arriving, and the vehicle base is growing. The commercial infrastructure (platforms, contracts, trust) does not yet exist at scale.

Why now is the window

Marketplaces are not won when the infrastructure is mature - they are won while it is still forming. The platform that has enrolled fleets, established partnerships, and built trust with vehicle owners before 2027 will be positioned to activate that capacity the moment the market is ready. The platform that waits for the rollout to be obvious will be too late.

Why It Does Not Fully Work Yet

Three structural tensions prevent V2G from working at scale today:

Hardware is ready on paper only

Most modern EVs contain an on-board charger (OBC) physically capable of bidirectional power flow. Most manufacturers have software-locked the feature. The first certified bidirectional wallboxes (Wallbox Quasar 2, Sigenenergy Sigenstor, Ambibox) only reached pilot deployment in 2025. ISO 15118-20 - the protocol that enables V2G communication - was published in 2022 but has almost no production deployment yet.

The chicken-and-egg problem.

To participate in the balancing market you need enrolled vehicles. To enrol vehicles you need to demonstrate market revenue. To access the market you need a BKV partner (see Section 4). BKV partners want to see an enrolled fleet first. Every piece depends on every other piece.

Data and trust.

A vehicle owner must trust a platform to remotely discharge an asset worth €15,000–50,000. The fleet operator knows parking patterns but does not understand energy markets. The platform needs both sides to function.

Purpose: This document gives you the industry context you need to understand the V2G problem space. It explains how the technology works, how the market is structured, who the actors are, and what the regulations say. It does not tell you what to build.

Read it as background, not as a blueprint.

1. What V2G Is

Electric vehicles spend most of their time parked. The average car is driven for roughly one hour per day; the other twenty-three hours it sits idle, connected to nothing or plugged into a charger. The battery during that time is doing nothing.

Vehicle-to-Grid (V2G) is the concept of using EV batteries during that idle time to provide services to the electricity system. Rather than treating the battery purely as a cost of ownership, V2G treats it as an asset that can participate in energy markets.

1.1 The distinction that matters: V1G vs V2G

Two terms are frequently confused:

	Smart charging (V1G)	Vehicle-to-Grid (V2G)
Direction of power flow	Grid to vehicle only	Grid to vehicle and vehicle to grid
What the vehicle does	Adjusts when it charges	Charges and discharges on demand
Market participation	Demand-side flexibility only	Can participate in energy supply markets
Hardware requirement	Standard charger with software	Bidirectional charger + compatible vehicle

Smart charging is a demand-side tool. V2G is a supply-and-demand tool. The distinction matters for which markets a platform can access and what services it can provide.

1.2 Related architectures: V2H and V2B

Two related concepts are relevant for understanding the full design space:

- **Vehicle-to-Home (V2H):** the EV battery powers the owner's home directly, bypassing the grid. No market participation or BKV required. Reduces the household's energy bill (Eigenverbrauch). Simpler to implement than grid-facing V2G but limited in scale.
- **Vehicle-to-Building (V2B):** same principle at building scale — a depot, office, or commercial facility uses its parked EV fleet to power the building during peak periods. Particularly relevant for fleet operators with on-site charging infrastructure.

V2H and V2B do not require balancing market participation or a BKV relationship. They can function as simpler first implementations or as parallel services alongside full V2G. The same physical hardware can support all three modes.

1.3 Service stacking

A single EV battery is not limited to one use case. In principle, the same battery can serve multiple purposes within a single day: charge from low-price solar generation at midday, power a building in the afternoon (V2B), and provide grid frequency response overnight

(V2G). This multi-use potential — sometimes called service stacking — is structurally different from any other energy storage asset class, where the battery is typically dedicated to one function.

Managing and optimising across these overlapping use cases is a core technical and commercial challenge.

1.4 Why EV batteries are structurally interesting for the grid

- The capital cost of the battery is already paid by the vehicle owner — for driving purposes. Grid services are an additional use of an asset already purchased.
- EVs are geographically distributed — parked in the same residential streets and commercial areas where local grid stress actually occurs.
- As EV adoption grows, the aggregate storage capacity available across a vehicle fleet becomes significant at national scale.

Important caveat

As of 2025, the number of vehicles in Germany that technically support bidirectional charging is small. Estimated in the tens of thousands. A significant portion have the feature software-locked by the manufacturer. The regulatory clock is currently moving faster than the hardware rollout.

2. How the Electricity Market Works

To understand where V2G fits, you need to understand the structure of electricity markets. Electricity cannot be stored at grid scale in conventional infrastructure — supply and demand must be balanced continuously, in real time. This creates a set of markets that pay for different types of flexibility.

2.1 The market layers

German electricity is traded across several time horizons. Each serves a different purpose:

Market	Time horizon	Purpose
Forward / futures market	Weeks to years ahead	Energy companies secure supply for future delivery. Prices are set far in advance.
Day-ahead market (EPEX Spot)	Next-day delivery; auction at 12:00	Hourly electricity prices for the following day are set at noon. Prices vary significantly — negative prices occur when renewable supply exceeds demand.
Intraday market	Hours ahead; continuous trading	Corrections to day-ahead positions as forecasts update. Particularly active around weather-driven generation changes.
Balancing / ancillary services market (Regelenergie)	Real-time; seconds to 30 minutes	Grid operators procure capacity that can respond instantly to maintain grid frequency at 50 Hz. This is the primary market where V2G assets are relevant.

2.2 Why grid frequency matters

The electricity grid operates at 50 Hz. If generation exceeds consumption, frequency rises. If consumption exceeds generation, it falls. Deviations beyond a narrow band damage equipment and can lead to outages. Grid operators must correct imbalances within seconds.

This creates a market for fast-responding assets. Parties who can inject or absorb power within seconds to minutes are paid for making that capacity available — regardless of whether they are actually called upon.

2.3 The balancing market in detail

Germany's four Transmission System Operators (TSOs) — 50Hertz, Amprion, TenneT, and TransnetBW — procure balancing services via the platform [regelleistung.net](https://www.regelleistung.net). There are three products, distinguished by response time:

Product	Response time	Activation mechanism	Payment structure	Minimum lot size
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FCR — Frequency Containment Reserve (Primärregelleistung)	< 30 seconds	Automatic — triggered directly by frequency deviation	Availability payment per MW per week; no separate payment per MWh delivered	1 MW (aggregated pools accepted)
aFRR — Automatic Frequency Restoration Reserve (Sekundärregelleistung)	< 5 minutes	Remote-controlled by the TSO	Availability payment plus energy payment when activated	1 MW (aggregated pools accepted)
mFRR — Manual Frequency Restoration Reserve (Tertiärregelleistung)	< 15 minutes	Manual instruction from TSO control room	Energy payment only when activated	1 MW (aggregated pools accepted)

Key structural point: the minimum lot size for all three products is 1 MW. A single EV with a 50 kW charger contributes 0.05 MW. Participation requires aggregating many vehicles into a single pool that behaves as one controllable unit.

2.4 Price behaviour

Balancing market prices are not fixed. FCR prices in Germany vary significantly by season, grid conditions, and the amount of competing capacity in the market. Prices are higher in winter when grid conditions are tighter, and lower in summer. Day-ahead and intraday prices exhibit strong patterns: periods of high renewable generation regularly produce low or negative prices, while peak demand periods produce high prices. The SMARD dataset (see Section 8) contains the full historical price series.

3. The Aggregation Structure

Because individual EVs cannot meet the 1 MW minimum lot size, participation requires an intermediate layer: an aggregator. Understanding this role is central to understanding how V2G businesses are structured.

3.1 What an aggregator does

An aggregator pools capacity from many vehicles or fleets and presents it as a single controllable block to the energy market. The aggregator acts as the interface between individual asset owners and the wholesale market. It handles market registration, bid submission, real-time dispatch, and settlement — distributing revenue back to contributing asset owners after taking a margin.

This model is not unique to V2G: wind farm aggregators, demand-response aggregators, and industrial flexibility aggregators operate on the same structural principle.

3.2 The battery economics question

A fundamental question for any V2G model is whether grid services are economically positive for vehicle owners after accounting for the cost of additional battery wear.

Every charge and discharge cycle degrades a battery — reducing its capacity over time. V2G services add cycling beyond normal driving patterns. Published research quantifies this additional degradation — the NREL/EPRI benchmark study (referenced in Section 8) provides the widely-cited numbers. The battery degradation datasets in your data pack give you the tools to model this per use pattern and cell chemistry.

Whether market revenue exceeds this degradation cost depends on the charger power level, the markets accessed, and how intensively the battery is used for grid services. At low charger power (AC, 7–22 kW), the economics are tight. At higher DC charger power (50–150 kW), more revenue is possible per vehicle.

For your analysis

Any credible product must account for battery degradation, not just market revenue. A V2G service that earns less than it costs vehicle owners in accelerated battery wear is not viable. The KIT and NREL/EPRI datasets in your data pack give you the numbers to model this properly.

4. The Balancing Responsible Party (BKV)

One of the most practically important – and least-discussed – structural features of the German energy market is the Bilanzkreisverantwortlicher (BKV), or Balancing Responsible Party. Any V2G market participant must understand this role.

4.1 What a Bilanzkreis is

The electricity grid operates on a settlement system in which every market participant must maintain a balanced position – the energy they inject into the grid must equal the energy they withdraw, within each 15-minute settlement period. This accounting unit is called a Bilanzkreis (balance circle).

The BKV is the entity legally responsible for keeping its balance circle in equilibrium. If the circle is out of balance, the TSO supplies corrective energy at penalty prices. The BKV trades on energy markets to forecast and manage their position – buying when they expect to be short, selling when they expect to be long.

4.2 Why V2G cannot reach the market without a BKV

A V2G aggregator that dispatches vehicles is, in market terms, injecting and withdrawing energy. All such activity must sit within a licensed Bilanzkreis. An aggregator that is not itself a licensed BKV must operate under the umbrella of an existing BKV – who assumes the settlement liability for the aggregator's positions.

Obtaining a BKV licence requires registration with the Bundesnetzagentur, technical certification, and proof of financial standing. The process typically takes 6–18 months. Most new market entrants therefore operate initially through a BKV partnership rather than seeking their own licence.

An important distinction: V2G participation is theoretically permitted under German law for any eligible technology. The gap between theoretically permitted and practically tradable is the BKV layer – along with the technical TSO pre-qualification and the commercial structure needed to manage settlement risk. This path from eligible to active market participant is a core part of the problem any platform in this space must address.

4.3 Who BKVs are

BKV type	Examples	Characteristics
Municipal utilities (Stadtwerke)	Stadtwerke München, Mainova, ENNI, and hundreds of smaller regional utilities	Regionally anchored; often interested in local flexibility assets; procurement decisions are slower but relationships are more durable
Large integrated utilities	E.ON, RWE, EnBW, Vattenfall	National scale; established trading desks; some are developing their own V2G products internally

Independent energy traders	Alpiq, Axpo, Statkraft Germany	Purely commercial orientation; interested in new tradeable assets; often more agile in negotiations
Specialist aggregators	The Mobility House (Germany), Jedlix (Netherlands, acquired by TenneT 2022)	Deep V2G expertise; natural partners or direct competitors depending on positioning

5. Regulatory Landscape

The regulatory environment for V2G in Germany and the EU is in active development. Some foundations are firmly established; others are being decided now; some critical questions remain open.

5.1 Established regulations

Regulation	In force since	What it establishes
AFIR – Alternative Fuels Infrastructure Regulation (EU-wide)	April 2024	Smart-charging capability is now mandatory at all new public charging points across the EU. All new public chargers must accept external energy management signals. This is the first regulatory layer that enables coordinated charging management at scale.
§ 14a EnWG – Controllable consumption devices (Germany)	2023	Grid operators may curtail EV charging during congestion periods. Vehicle owners who register receive a reduced network tariff in exchange. Establishes the legal basis for remote controllability of EV charging – a prerequisite for any V2G dispatch.
ISO 15118-2 – Smart charging communication mandate (EU-wide)	January 2026	All new AC public charging infrastructure must support ISO 15118-2, enabling Plug & Charge authentication and smart charging communication. This is the communication foundation layer below V2G.

5.2 Upcoming regulations

Regulation	Effective date	What it requires
ISO 15118-20 – Bidirectional charging mandate (EU-wide, all new public and private chargers)	January 2027	All new charging infrastructure – both public and private – must support ISO 15118-20, the standard that enables true bidirectional power flow. From this date, V2G communication capability becomes a baseline requirement for new hardware installations across the EU. This is the most significant near-term regulatory milestone for V2G.
EU Battery Regulation – State of Health certification	2030 (phased)	EV batteries must carry certified State of Health documentation. This has implications for V2G: documented degradation caused by grid services creates accountability questions between vehicle owners, aggregators, and manufacturers.

5.3 Open regulatory questions

The following issues are unresolved as of mid-2025. They represent genuine regulatory risk that any V2G business model must account for:

Open question	Why it matters
Network charges on reverse power flow (§ 19 StromNEV)	It is not definitively settled whether an EV owner feeding electricity back through a residential meter incurs network charges. If they do, the economics of residential V2G are significantly affected.
Tax and VAT status of V2G income for private individuals	If private vehicle owners receiving V2G payments are classified as commercial energy producers, they may face VAT registration obligations. Current interpretation suggests aggregator structures may avoid this, but it is not tested.
Marktstammdatenregister (MaStR) registration unit	Any party feeding electricity to the grid must register in the Market Master Data Register. Whether individual vehicles or the aggregator as a collective unit is the registerable entity is not yet resolved.
Liability for V2G-induced battery degradation	The EU Battery Regulation creates SoH accountability, but does not specify who bears responsibility for above-normal degradation attributable to grid services.

6. The Technology Stack

V2G is not a single technology but a stack of interdependent layers. Every layer must function correctly – a problem at any level prevents the system from working.

6.1 Layer overview

Layer	Component	Function	Current state (2025)
1	Battery	Electrochemical storage. Battery chemistry affects how many charge/discharge cycles can be performed before significant capacity loss.	LFP chemistry (used in BYD and some other manufacturers) tolerates more cycling than NMC chemistry (more common in European premium vehicles). Chemistry selection is relevant for V2G intensity.
2	On-board charger (OBC)	The converter inside the vehicle that transforms AC grid power to DC for the battery – and in a bidirectional vehicle, reverses that process. Must be designed and software-enabled for bidirectionality.	This is the primary hardware constraint. Many vehicles contain OBC hardware capable of bidirectionality but with the feature software-locked by the manufacturer. See Section 6.3.
3	Charging hardware (EVSE)	The physical charger connecting vehicle to grid. Must support bidirectional power flow for V2G. AC bidirectional (3.7–22 kW) is slower but cheaper. DC bidirectional (50–150 kW) is faster but significantly more expensive.	The first CCS bidirectional wallboxes – Wallbox Quasar 2, Sigenenergy Sigenstor, and Ambibox – reached pilot deployment in 2025. Broader availability expected 2026–2027.
4	Communication protocol	The standard by which vehicle and charger exchange information: authentication, energy scheduling, bidirectional power commands. ISO 15118-20 is the relevant V2G standard.	ISO 15118-20 was published in 2022. Production vehicles with active bidirectional software are still rare – most hardware currently uses ISO 15118-2 (smart charging only) or proprietary protocols.
5	Platform and market connectivity	Software that aggregates vehicle states, forecasts availability, interfaces with energy markets, dispatches vehicles, and accounts for revenue and battery usage.	This is the software layer. It requires real-time data from vehicles and chargers, market API connectivity, and optimisation logic.

6.2 Communication protocols

Standard	What it enables	Market status
ISO 15118-2	Plug & Charge (automatic vehicle authentication); smart charging scheduling; one-way power management between vehicle and charger	Mandatory for new AC charging infrastructure in the EU from January 2026. Widely available in 2025 hardware. Reference: iso.org/standard/55366
ISO 15118-20	Everything in ISO 15118-2, plus: bidirectional power flow, external dispatch commands for V2G and V2H/V2B services, dynamic energy service scheduling	Published 2022. Mandatory for all new EU charging infrastructure from January 2027. Currently in early pilot deployment only.
OCPP 2.0.1	Backend protocol connecting chargers to management platforms. OCPP 2.0.1 added support for bidirectional energy management. Earlier versions (1.6, 2.0) did not.	Available but not universally deployed. A V2G platform requires OCPP 2.0.1 support from connected hardware.

6.3 Vehicle compatibility – current status

An important distinction: having ISO 15118-20 hardware in a vehicle does not mean V2G is active. Many manufacturers ship vehicles with the bidirectional OBC hardware in place but the feature disabled in software. Activation requires a manufacturer decision — sometimes a paid software update, sometimes a market-specific regulatory trigger. The Kia EV9 is an illustrative example: prototype versions have demonstrated V2G over ISO 15118-20 in trials, but production EV9s ship without the bidirectional feature activated.

Vehicle	V2G capability	Connector	Notes
Nissan Leaf (2013–2023)	Active in pilots	CHAdeMO	V2G pioneer. Well-documented in research. CHAdeMO connector standard is being phased out with minimal new infrastructure deployment.
Mitsubishi Outlander PHEV	Active in pilots	CHAdeMO	Same connector situation as Leaf.
Hyundai IONIQ 5 / Kia EV6 (2022+)	Active, expanding	CCS / ISO 15118-20	First mainstream vehicles using CCS-based V2G. Activated in select European markets.
Kia EV9	Prototype demonstrated, series locked	CCS / ISO 15118-20	V2G demonstrated in trials. Production vehicles do not have bidirectional feature activated as of mid-2025.
BYD Atto 3, Han, Tang	Partial — market-dependent	CCS	Bidirectional hardware present. Software availability varies by market.

Volkswagen ID.3, ID.4, ID.7	Hardware-ready, software-locked	CCS	VW has communicated intention to enable V2G via software update. Timeline not confirmed as of mid-2025.
BMW iX, i4, i5 (2024+)	In development	CCS	Bidirectional capability announced. Limited activation as of mid-2025.
Tesla (all models)	Not supported	NACS / CCS adapter	Tesla has not made V2G available on any model.

7. Market Actors

The V2G market involves multiple actor types with different roles, interests, and technical capabilities.

Actor	Role in the system	What they care about
Transmission System Operators (TSOs) 50Hertz, Amprion, TenneT, TransnetBW	Operate high-voltage transmission network; procure balancing services (FCR, aFRR, mFRR) to maintain grid frequency	Reliable, fast-responding capacity at lowest cost; as thermal plants retire and renewables grow, need for distributed flexible assets increases
Distribution System Operators (DSOs) e.g. Bayernwerk, E.ON Netz, Stromnetz Berlin	Operate local distribution networks (the wires to homes, offices, industrial sites)	Preventing congestion in local networks, especially in areas with high EV adoption; deferring costly grid reinforcement investments
Balancing Responsible Parties (BKV) Stadtwerke, energy traders	Hold licensed Bilanzkreise; trade energy to maintain balance; can host third-party aggregators under their licence	Accessing new flexible assets; revenue from ancillary service markets; managing balance circle risk
Fleet operators Car-sharing, leasing, logistics, municipalities, corporates	Operate multiple EVs; responsible for vehicle availability, charging logistics, total cost of ownership	Reducing total cost of operating EVs; integrating energy services with fleet management; meeting sustainability targets
Individual vehicle owners	Own or lease a privately used EV; vehicle is parked 16–22 hours per day on average	Passive additional income from parked vehicle; energy bill reduction; minimal additional complexity
Charging hardware manufacturers ABB, Wallbox, Alfen, Kempower, Heliox	Design and manufacture EVSE; increasingly developing V2G-capable products	Product differentiation; software and services revenue; enterprise fleet contracts
Vehicle manufacturers (OEMs) Hyundai, BYD, VW, BMW, Renault	Design and manufacture EVs; control software enablement of bidirectional OBC capability	Customer retention; new revenue streams from connected vehicles; EU V2G readiness compliance

8. Dataset Reference

The following datasets are available for this challenge. This section explains what each one contains, where to find it, and what kind of questions it can help you answer. Use what you need or look for your own data.

8,1 Challenge data

OPEN

Real Grid Data: SMARD.de (Bundesnetzagentur)

Download <https://smard.de/home/downloadcenter/download-marktdaten>

Official German electricity market data from the Bundesnetzagentur. Contains 15-minute resolution data for: electricity generation by source (solar, wind, gas, nuclear, etc.), total national consumption, import/export flows, and day-ahead spot prices. Data runs from 2015 to present. Free download, no account required.

Use this to understand when electricity is cheap (high renewable periods, negative prices) and when it is expensive (peak demand). This is the demand side of the V2G equation: what the grid pays for flexibility at any given moment.

Dummy

Synthetic Dataset: Fleet Charging Sessions

Approximately 8,000 synthetic EV charging sessions generated over a 6-month period, representing a fleet operating in the Rhein-Main region. Fields include: timestamps (start/end), location type, postal code, kWh delivered, battery capacity, and State of Charge at session start and end.

This represents the supply side of a V2G marketplace: what a fleet's available battery capacity looks like over time, where vehicles are when they are parked, and how much energy is available at each point. Use it to model availability patterns, geographic distribution, and the aggregate pool that could be offered to the market.

OPEN

Regelleistung.net — German Balancing Energy Market

Portal <https://regelleistung.net>

Germany's balancing energy market portal, operated jointly by the four TSOs. Contains historical FCR, aFRR, and mFRR tender results: volumes tendered, clearing prices, awarded capacity, and market rules. Weekly granularity.

This is where flexibility is actually traded and priced. Use it to understand what the buyer side of the V2G market looks like: what capacity prices have been, how they vary seasonally, and how the procurement process works.

OPEN

ENTSO-E Transparency Platform

Portal — <https://transparency.entsoe.eu>

Pan-European grid data: generation by source, total load, cross-border flows, and balancing volumes. Free API access. Covers all EU member states.

Useful for understanding how the German grid interacts with neighbouring countries and for contextualising German grid stress events within the broader European picture.

OPEN

Kraftfahrtbundesamt (KBA) – EV Registration StatisticsPortal – <https://kba.de>

Official German federal vehicle authority statistics. Monthly new EV registrations broken down by vehicle model, manufacturer, fuel type, and region. The most authoritative source for EV fleet composition data in Germany.

Use this to understand which vehicle models are entering the German fleet, at what rate, and where they are concentrated. Relevant for sizing the potential V2G-capable vehicle base by geography and model.

OPEN

Bundesnetzagentur LadesäulenregisterDownload – <https://www.bundesnetzagentur.de/DE/Fachthemen/ElektrizitaetundGas/E-Mobilitaet/DownloadundKontakt.html>

The official register of all reported public charging points in Germany: location, connector type, power rating, and operator. Free CSV download, updated regularly.

Use this to map where charging infrastructure currently exists, what connector types are deployed, and what power levels are available. Relevant for understanding where bidirectional-capable hardware is actually installed.

8.2 Battery degradation datasets

Modelling how much battery capacity is safely available for V2G – without degrading the battery beyond acceptable limits – requires degradation data. Three open datasets are available:

OPEN

Battery ArchivePortal – <https://batteryarchive.org>

A central aggregator for open battery cycling data from multiple research sources. Includes the Sandia National Laboratories dataset of 86 commercial 18650 cells across three chemistries (NMC, NCA, LFP), cycled at varying Depth of Discharge windows (0–100%, 20–80%, 40–60%). Unified CSV schema across sources. Free download.

Use this to understand how different cell chemistries degrade under different cycling patterns, and how limiting the depth of discharge affects cycle life. Directly applicable to modelling V2G cycling strategies.

OPEN

KIT V2G Aging DatasetDOI: 10.35097/1947 | Python model – <https://github.com/energystatusdata/bat-age-model>

Designed specifically to compare battery aging under three conditions: standard EV charging, smart charging (V1G), and V2G cycling. Uses NMC/C-SiO cells and tracks degradation past the "knee point" – the inflection where capacity loss accelerates – down to 40–50% remaining capacity. A ready-to-use Python simulation model is available at the GitHub link above.

The most directly relevant dataset for V2G degradation analysis. Use it to quantify how much additional capacity loss V2G cycling causes compared to standard EV use, and at what cycling intensity the degradation cost outweighs the market revenue.

OPEN

NREL/EPRI V2G Impact Benchmark Study

Report – <https://epri.com/research/products/000000003002024770>

Not raw data, but the peer-reviewed reference numbers that anchor any serious V2G economics analysis: EV-only battery degradation runs at approximately 1.5% capacity loss per year at 30°C. Daily V2G use at 43% battery energy turnover adds approximately 1.8% per year, leading to approximately 33% total capacity fade over 10 years versus approximately 15% for EV-only use.

Use these benchmark numbers as the basis for your degradation cost calculations. When making claims about battery economics, cite this study – it is the standard reference in industry and academic V2G literature.

OPEN

NASA PCoE Randomized Battery Usage Dataset

Portal – <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>

NASA Prognostics Center of Excellence dataset covering randomised charge/discharge profiles on Li-ion cells. Useful as an additional reference for degradation modelling under irregular cycling patterns.

Fallback dataset if the KIT and Battery Archive data do not cover a specific scenario you need to model.

8.3 Weather and renewable generation data

V2G dispatch opportunity correlates with grid conditions, which in turn correlate with renewable generation and weather patterns.

OPEN

DWD – Deutscher Wetterdienst Open Data

Portal – <https://opendata.dwd.de>

Historical and forecast German weather data from the national meteorological service. Includes solar irradiance, wind speed, and temperature at station and grid level. Free API, no registration required.

Use this to correlate solar and wind generation patterns with electricity price data from SMARD. Useful for understanding when low-price charging windows occur and for building forward-looking availability models.

OPEN

Open-Meteo API

Portal – <https://open-meteo.com>

Free global weather forecast API with a 16-day forecast horizon. No API key or registration required.

Lightweight alternative to DWD for building forecasting models that anticipate dispatch opportunities based on upcoming weather conditions.